Symbolic circuit reference: selected identities and comparisons

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This appendix records exact identities and scoped comparisons. The revised manuscript and its current independent compiler table are authoritative. The atlas is not claimed to dominate every tool or publication.

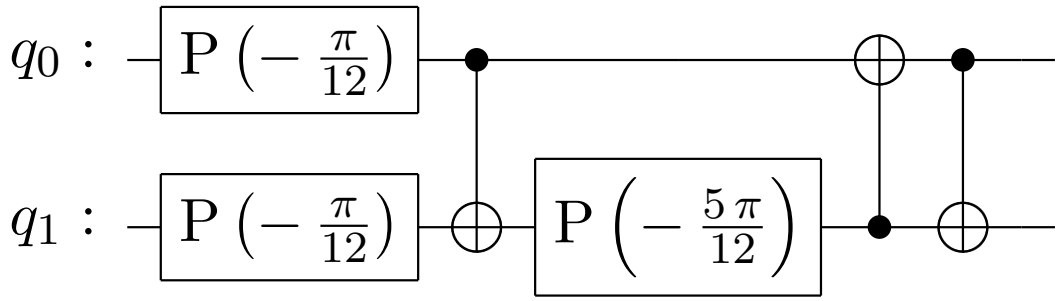
1 Controlled fermionic SWAP

The literal CX/U3 circuit has four CNOTs and six locals. The interaction count is optimal among arbitrary Ising-class interactions without ancillas; the local count is an upper bound. BQSKit independently finds four-CNOT, six-local circuits, so discovering the upper bound is not a new compiler advantage. The lower theorem extends to all controlled balanced involutions. The three-qubit hardware family of Warren et al. is related physical context, not a proof of these CNOT counts [1].

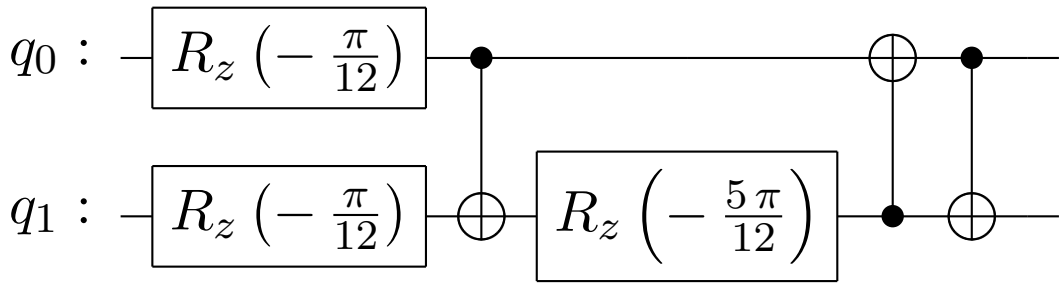
2 Sycamore phase contract

For SYC = fSim($\pi/2, \pi/6$), the exact P/CX circuit and the projective Rz/CX circuit below each use three entanglers and three locals. In the Rz circuit, SYC = $e^{-7i\pi/24}C$. The phase-polynomial certificate proves the three-phase lower bound only in the CNOT/P subalphabet; it does not establish an unrestricted local minimum. The former tomography-figure comparison has been withdrawn.

cx_h_phase



ibm_cx



3 Current independent fixed-target comparison

The table gives E/L in CX/U3 and the best verified recorded route for each tool. Inputs are independent library words or target matrices, not the atlas output. A dash excludes all recorded outputs on that route that fail its replay threshold or budget. Quartz and QUESO optimize total NAM gates; their locals are then fused to U3 without changing the entangling skeleton.

Gate	Qiskit	tket	BQSKit	Quartz	QUESO	Exact atlas
SYC	3/8	3/7	3/4	4/5	4/5	3/3
iSWAP	2/3	2/4	2/4	2/3	2/3	2/3
SSWAP	3/8	3/7	3/5	–	3/8	3/7
fFredkin	5/13	10/15	4/6	9/10	13/10	4/6
CCX	6/8	6/8	6/8	6/8	6/8	6/8
CSWAP	7/10	7/8	7/9	8/8	8/8	7/10
C3X	14/19	14/16	14/16	14/16	14/16	14/17
C4X	36/41	36/41	–	30/33	36/36	36/41

BQSKit matches fFredkin and beats the SSWAP local count. tket and BQSKit beat the C3X local count. Quartz's 30-CX C4X circuit beats the current 36-CX atlas entry. These losses are retained. The earlier eleven SDK entangler wins were all fFredkin in different alphabets.

4 Historical broad local-count audit

The CSV files retain comparisons against the older, explicitly identified SDK routes, including source-fed routes. They are not comparisons against all compiler strategies. Counts and parameter-description score are different objectives; a smaller L alone does not prove a better archived score.

Historical status	Cells
equal_E_equal_locals	2108
equal_E_fewer_locals	553
equal_E_more_locals	1
fewer_entanglers	11

5 Established native bounds and proof scope

The universal two-B construction [2] and the SQiSW two-pulse Weyl region and three-pulse universal construction [3] are prior results. The released exact charts implement them with explicit scalar handling and branch definitions. The new invariant iff proof rewrites the known SQiSW region;

it is not a new universal pulse cap. The 192 one-qubit phase-resolved Clifford circuits are a finite reference, not a large unexplored class.

6 Reproduction

The companion JSON and CSVs retain archive hashes, exact identities, older comparison settings, and every local-count regression. The revised review bundle additionally includes the stronger SDK/Quartz/QUESO outputs, failures, optimality ledger, theorem checks, and both complete atlas PDFs and sources. UniversalQCompiler has no executed comparison because its licensed kernel was unavailable. No numerical sample proves a symbolic family identity or an unrestricted global minimum.

References

[1] C. W. Warren, J. Fernández-Pendás, S. Ahmed, et al., Extensive characterization and implementation of a family of three-qubit gates at the coherence limit, npj Quantum Information 9, 44 (2023), doi:10.1038/s41534-023-00711-x.

[2] J. Zhang, J. Vala, S. Sastry, and K. B. Whaley, Minimum construction of two-qubit quantum operations, Physical Review Letters 93, 020502 (2004), arXiv:quant-ph/0312193.

[3] C. Huang et al., Quantum instruction set design for performance, Physical Review Letters 130, 070601 (2023), arXiv:2105.06074.